YELLOW REPORT

Gabapentin should be reserved for **symptom control** (pain, neuropathic symptoms, or seizure prophylaxis) [1] in this patient, not for overall survival. There is **no high-quality evidence** that gabapentin improves survival in glioblastoma, and any potential survival benefit is unproven and outweighed by risks in this frail, ECOG 3 patient with recent falls and cognitive impairment [2]. If gabapentin is used, start at **100 mg three times daily** and titrate slowly to 300 mg three times daily, monitoring closely for sedation, confusion, and fall risk; **avoid high doses** and discontinue if intolerable. Given his ECOG 3 status, recent fall, and cognitive impairment, **prioritize fall-prevention and nonpharmacologic strategies** [3] for symptoms, and avoid gabapentin for survival benefit.

Evidence for gabapentin in glioblastoma

- **Survival benefit**: A 2025 retrospective cohort study reported improved survival with postoperative gabapentin, but this is not conclusive evidence of a survival benefit; interpretation should be limited by the retrospective, observational design with potential for residual confounding and immortal time bias, and randomized confirmation is awaited [2].
- **Mechanism**: Gabapentin may modulate neuron-tumor interactions via α2δ-1, but this is speculative and not supported by prospective clinical data [2].
- **Guidelines**: No major neuro-oncology guidelines recommend gabapentin for survival benefit in glioblastoma.

Clinical considerations for this patient

Given his frailty and comorbidities, **key considerations** include:

- **Performance status**: ECOG 3 indicates severe limitations; high fall risk and cognitive impairment make sedation and confusion especially hazardous.
- **Fall risk**: Recent falls and a shuffling gait heighten the risk of gabapentin-related injuries.
- **Cognitive impairment**: Gabapentin can worsen confusion and memory deficits, especially in older adults.

Recommended dosing and monitoring

If gabapentin is used for symptom control, **start low and titrate slowly**:

Parameter	Recommendation
Initial dose	100 mg three times daily
Titration	Increase by 100 mg three times daily every 3–7 days as tolerated
Target dose	300 mg three times daily (max 900 mg/day)
Monitoring	Sedation, confusion, gait instability, falls, and medication interactions
Discontinuation	Stop if intolerable or no benefit after 2–4 weeks

Risks and benefits assessment

- **Benefits**: Potential modest symptomatic relief (e.g. neuropathic pain).
- **Risks**: Sedation, confusion, falls, and medication interactions are significant in this patient.
- **Balance**: Risks likely outweigh unproven survival benefits; use only if symptoms are refractory and monitored closely.

Alternative strategies

Prioritize **nonpharmacologic and symptom-focused care**:

- **Fall prevention**: Physical therapy, home safety, and caregiver support.
- **Symptom management**: Tailored analgesia, seizure prophylaxis if indicated, and supportive care [1].
- **Shared decision-making**: Align treatment with patient values and goals of care.

Sources:

1. SNO and EANO practice guideline update: anticonvulsant prophylaxis in patients with newly diagnosed brain tumors. Neuro-Oncology, 2021
2. Gabapentinoids confer survival benefit in human glioblastoma. Nature Communications, 2025
3. European association for neuro-oncology (EANO) guidelines for palliative care in adults with glioma. The Lancet: Oncology, 2017